# Syllabus 9758

Statistics Summary

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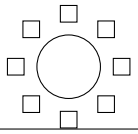
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# 1 Permutations and Combinations

## 1.1 Counting Principles and Formulae

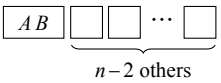
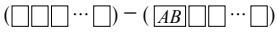
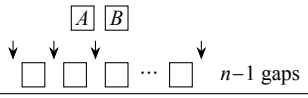
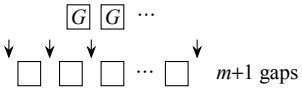
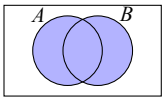
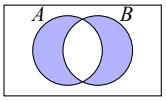
| | Addition principle | Multiplication principle |
|---|---|---|
| Use when | either event $A$ **or** $B$ (cases) | stage 1 **and** stage 2 (steps) |
| Condition | cases cannot happen together | stages are done one after another |
| Total | $m+n$ | $m\times n$ |

| Situation | Formula | Visual | Note |
|---|---|---|---|
| $n$ distinct objects in a row | $n!$ | $\boxed{n}\boxed{n-1}\boxed{n-2}\cdots\boxed{1}$ | $0!=1$ |
| $r$ of $n$ distinct objects, in order | ${}^nC_r\times r!=\dfrac{n!}{(n-r)!}$ | $\boxed{n}\boxed{n-1}\cdots\boxed{n-r+1}$ | order matters |
| $r$ of $n$ distinct objects, no order | ${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$ | $\underbrace{\bullet\bullet\bullet\bullet}_{\text{pick } r}\bullet\bullet\bullet\cdots\bullet$ | ${}^nC_r={}^nC_{n-r}$ |
| $r$ places, each of $n$ choices, repeats allowed | $n^r$ | $\boxed{n}\boxed{n}\boxed{n}\cdots\boxed{n}$ | objects **can** be reused |
| $n$ objects, $p$ alike of one kind, $q$ alike of another | $\dfrac{n!}{p!\,q!}$ | $\underbrace{\boxed{a}\cdots\boxed{a}}_{p\text{ times}}\underbrace{\boxed{b}\cdots\boxed{b}}_{q\text{ times}}$ | divide by $k!$ for each $k$ identical |
| $n$ distinct objects in a circle | $\dfrac{n!}{n}=(n-1)!$ |  | rotations are the same |

**GC:** `nPr`, `nCr`: `MATH` → `PRB`. Type $n$ first, then the command, then $r$. `!` is in the same menu.

> **Note**
>
> Order matters ⇒ select then arrange: ${}^nC_r\times r!$

# 1.2 Arrangements in a Line

| Condition | Method | Count | Visual |
|---|---|---|---|
| Fixed position (e.g. $A$ first; number is even; first digit $\neq 0$) | fill the restricted places **first**, then the rest | $\underline{k} \times _ \times \cdots$ |  |
| $A$, $B$ **together** | Grouping method: group $A$ and $B$ together, then arrange the blocks; arrange inside the block. | $(n-1)! \times 2!$ |   |
| $A$, $B$ **separated** (complement) | total − together | $n! - (n-1)!\,2!$ |  |
| $A$, $B$ **separated** (slot in) | slot in: arrange the other $n-2$; choose 2 of the $n-1$ gaps for $A$, $B$; arrange $A$, $B$ | $(n-2)! \times {}^{n-1}\mathrm{C}_2 \times 2!$ |   |
| No two of a group adjacent | **slot in method**: arrange the other $m$; choose $k$ of the $m+1$ gaps for the group; arrange the group | $m! \times {}^{m+1}\mathrm{C}_k \times k!$ |   |
| "at least one" | complement: total − none | | |
| $A$ **or** $B$ (or both) | add, then subtract the overlap once | $n(A)+n(B)-n(A\cap B)$ |   |
| $A$ **or** $B$ but not both | add, then subtract the overlap twice | $n(A)+n(B)-2n(A\cap B)$ |   |
| Several cases (e.g. last digit 0 or not) | case table: count each case, then **add** | | |

**Example 1.1**

Find the number of arrangements of the letters in the word STATISTICS.

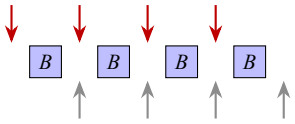
$$\text{S}\times 3,\ \text{T}\times 3,\ \text{I}\times 2$$

$$\text{total} = \frac{10!}{3!\,3!\,2!}$$

$$= 50\,400$$

**Example 1.2**

Find the number of ways 4 boys and 4 girls can be arranged in a line such that boys and girls alternate.

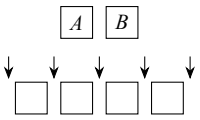



$$\text{total} = 2\times 4!\times 4!$$

$$= 1152$$

**Example 1.3**

How many ways are there to arrange 6 people in a line with $A$ and $B$ separated?

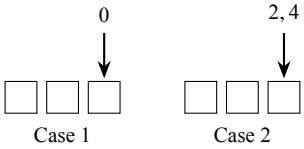



$$\text{total} = 4!\times {}^5\text{C}_2\times 2!$$

$$= 480$$

**Example 1.4**

Even 3-digit numbers from 0, 1, 2, 3, 4, 5, no repeats



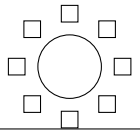
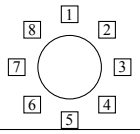
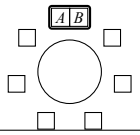
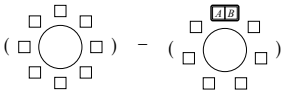
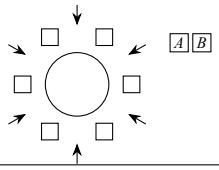
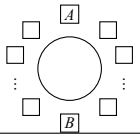
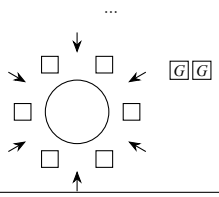

$$\text{last } 0 = 5\times 4 = 20$$

$$\text{last } 2 \text{ or } 4 = 2\times 4\times 4 = 32$$

$$\therefore \text{total} = 20+32$$

$$= 52$$

# 1.3 Circular Arrangements

| Situation | Method | Count | Visual |
|---|---|---|---|
| $n$ people round a table | | $(n-1)!$ |  |
| seats **numbered / distinguishable** | ignore the numbered seats and do everything as per normal, then multiply by the number of seats at the end | $(n-1)!\times n$ |  |
| $A$, $B$ **together** | grouping, then circular arrangement of the blocks | $(n-2)!\times 2!$ |  |
| $A$, $B$ **separated** (complement) | total − together | $(n-1)!-(n-2)!\times 2!$ |  |
| $A$, $B$ **separated** (slot in) | slot in: arrange the other $n-2$ in a circle; choose 2 of the $n-2$ gaps for $A$, $B$; arrange $A$, $B$ | $(n-3)!\times {}^{n-2}\mathrm{C}_2\times 2!$ |  |
| $A$, $B$ in fixed relation (e.g. $A$ opposite $B$) | fix $A$; $B$ has 1 place; arrange the rest | $(n-2)!$ |  |
| no two of $k$ people adjacent | **slot in method**: arrange the other $m$ in a circle; choose $k$ of the $m$ gaps; arrange the $k$ people | $(m-1)!\times {}^{m}\mathrm{C}_k\times k!$ |  |

## 1.4 Selections and Groups

| Situation | Method | Count |
|---|---|---|
| choose $r$ of $n$, with a condition (e.g. $\geq 1$ woman) | ${}^nC_r$; cases (add), or complement | total − none |
| choose, **then** arrange | select, then multiply by $r!$ (or by the arrangements) | ${}^nC_r \times r!$ |
| "at least one of **each** kind" | consider cases<br>Do **not** choose one of each first, then mix the rest together and choose the remaining (overcounts) | add cases |
| forming words / codes with **repeated** letters | cases by the pattern of repeats (all different; one pair; …): choose the letters, then arrange $\frac{r!}{p!\,q!}$ | add the cases |
| groups of **different** sizes / labelled groups | choose group by group; named teams, different tasks or sizes: **no** division | ${}^nC_a \, {}^{n-a}C_b \cdots$ |
| $k$ groups of the **same** size, unlabelled | as above, then $\div k!$ (identical groups of equal size) | $\frac{{}^nC_a \, {}^{n-a}C_a \cdots}{k!}$ |

**Example 1.5**

8 people into two groups of 4

$$\text{unlabelled} = \frac{{}^8C_4}{2!} = 35$$

$$\text{teams } X, Y = {}^8C_4 = 70$$

**Example 1.6**

3-letter codes from MISSISSIPPI

$$\text{all different} = {}^4C_3 \times 3! = 24$$

$$\text{one pair} = (3 \times 3) \times \frac{3!}{2!} = 27$$

$$\text{all alike} = 2$$

$$\therefore \text{total} = 53$$

**Example 1.7**

5 from 3 red, 4 green, 5 blue; every colour used

$$(3, 1, 1) = {}^3C_3 \times {}^4C_1 \times {}^5C_1 = 20$$

$$(1, 3, 1) = {}^3C_1 \times {}^4C_3 \times {}^5C_1 = 60$$

$$(1, 1, 3) = {}^3C_1 \times {}^4C_1 \times {}^5C_3 = 120$$

$$(2, 2, 1) = {}^3C_2 \times {}^4C_2 \times {}^5C_1 = 90$$

$$(2, 1, 2) = {}^3C_2 \times {}^4C_1 \times {}^5C_2 = 120$$

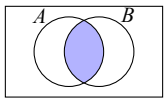
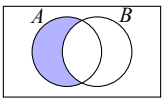
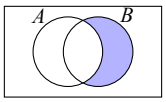
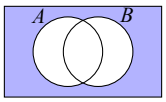
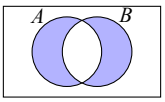
$$(1, 2, 2) = {}^3C_1 \times {}^4C_2 \times {}^5C_2 = 180$$

$$\therefore \text{ways} = 20 + 60 + 120 + 90 + 120 + 180 = 590$$

# 2 Probability

## 2.1 Basic Rules

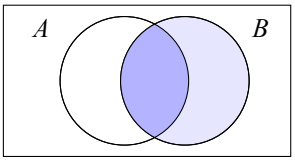
| Result | Formula | Note |
|---|---|---|
| Equally likely outcomes | $\mathrm{P}(A)=\frac{n(A)}{n(S)}$ | $0\leqslant \mathrm{P}(A)\leqslant 1$ |
| Complement | $\mathrm{P}(A')=1-\mathrm{P}(A)$ | "at least one" = 1 − none |
| Addition rule | $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $A\cup B$: $A$ or $B$ (or both) |
| Mutually exclusive | $\mathrm{P}(A\cap B)=0$ | so $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Exhaustive | $\mathrm{P}(A\cup B)=1$ | |

|  $A\cap B$ |  $A\cap B'$ |  $A'\cap B$ |  $A'\cap B'=(A\cup B)'$ |  exactly one |
|---|---|---|---|---|
| $\mathrm{P}(A\cap B)$ | $\mathrm{P}(A)-\mathrm{P}(A\cap B)$ | $\mathrm{P}(B)-\mathrm{P}(A\cap B)$ | $1-\mathrm{P}(A\cup B)$ | $\mathrm{P}(A)+\mathrm{P}(B)-2\mathrm{P}(A\cap B)$ |

The four regions add up to 1. $(A\cup B)'=A'\cap B'$ and $(A\cap B)'=A'\cup B'$. None of these probability results is in MF27.

## 2.2 Conditional Probability

| Result | Formula | Note |
|---|---|---|
| Conditional probability | $\mathrm{P}(A\mid B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$ | probability of $A$ given $B$ has happened; $\mathrm{P}(B)\neq 0$ |
| Multiplication rule | $\mathrm{P}(A\cap B)=\mathrm{P}(A\mid B)\mathrm{P}(B)=\mathrm{P}(B\mid A)\mathrm{P}(A)$ | used along a tree branch |
| Complement given $B$ | $\mathrm{P}(A'\mid B)=1-\mathrm{P}(A\mid B)$ | |
| Warning | $\mathrm{P}(A\mid B)\neq \mathrm{P}(B\mid A)$ | different denominators |

reduced sample space: 


$$\mathrm{P}(A\mid B)=\frac{\text{shaded}}{\text{whole circle } B}$$

## 2.3 Independent and Mutually Exclusive Events

| | Independent | Mutually exclusive |
|---|---|---|
| Meaning | one event does not change the probability of the other | cannot both happen |
| Test | • $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$<br>• $\mathrm{P}(A\mid B)=\mathrm{P}(A)$<br>• $\mathrm{P}(B\mid A)=\mathrm{P}(B)$ | • $\mathrm{P}(A\cap B)=0$<br>• $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$ |
| Venn | circles overlap, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ | circles do not overlap |

**Note**

- If $A$, $B$ are independent, so are $A$ and $B'$, $A'$ and $B$, $A'$ and $B'$.
- If $\mathrm{P}(A)>0$ and $\mathrm{P}(B)>0$, events cannot be both independent and mutually exclusive: $\mathrm{P}(A\cap B)=0\neq\mathrm{P}(A)\mathrm{P}(B)$.
- To show independence, show $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$; *do not* assume it from the wording.
- Independence cannot be detected from the shape of the Venn diagram: it generally looks the same as a Venn diagram in which $A$ and $B$ are not independent.

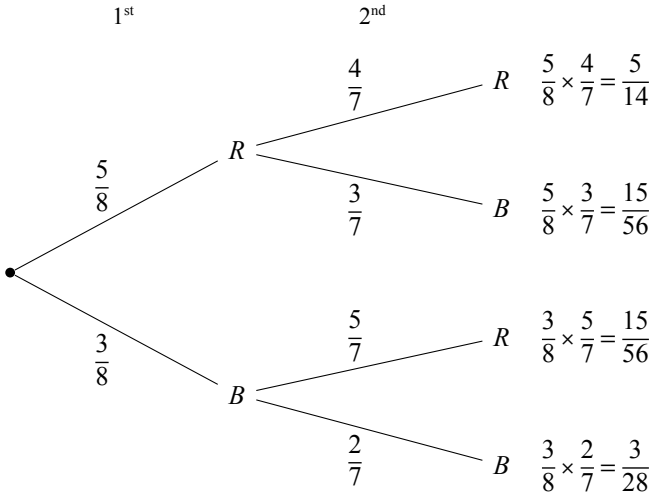
**Example 2.1**

Given that $A$ is independent of $B$, prove that $A'$ is independent of $B'$.

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A\cup B)\\ &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A\cap B)\\ \because\ \mathrm{P}(A\cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\ \therefore\ \mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A)\mathrm{P}(B)\\ &= (1-\mathrm{P}(A))(1-\mathrm{P}(B))\\ &= \mathrm{P}(A')\mathrm{P}(B')\\ \therefore\ A',\ B' &\text{ are independent}\end{aligned}$$

## 2.4 Tree Diagrams

Multiply along branches (∩); add across leaves (∪). Branches from one node sum to 1. Without replacement: probabilities in the second stage **change**.




Bag: 5 red, 3 blue; 2 drawn without replacement.

$$\begin{aligned}\mathrm{P}(\text{different colours}) &= \frac{5}{8}\times\frac{3}{7}+\frac{3}{8}\times\frac{5}{7}\\ &= \frac{15}{56}+\frac{15}{56}\\ &= \frac{15}{28}\end{aligned}$$

$$\begin{aligned}\mathrm{P}(1\text{st } R \mid 2\text{nd } B) &= \frac{\mathrm{P}(1\text{st } R \text{ and } 2\text{nd } B)}{\mathrm{P}(2\text{nd } B)}\\ &= \frac{\mathrm{P}(R,B)}{\mathrm{P}(R,B \text{ or } B,B)}\\ &= \frac{\frac{15}{56}}{\frac{15}{56}+\frac{6}{56}}\\ &= \frac{5}{7}\end{aligned}$$

## 2.5 Tables of Outcomes and Counting Methods

| + | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |

two dice: $n(S)=36$

$$\mathrm{P}(\text{sum}=7)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}\geqslant 10)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}=7 \mid \text{a 3 shown})=\frac{2}{11}$$

**Example 2.2**

There are 6 men and 10 women. 5 people are chosen. Find the probability that the group has 3 men.

(a)

$$\begin{aligned}\mathrm{P}(3M, 2W) &= \frac{6}{16}\times\frac{5}{15}\times\frac{4}{14}\times\frac{10}{13}\times\frac{9}{12}\times\frac{5!}{2!\,3!}\\ &= \frac{15}{728}\times 10\\ &= \frac{75}{364}\end{aligned}$$

(b)

$$\begin{aligned}\mathrm{P}(3 \text{ men}) &= \frac{{}^{6}\mathrm{C}_3\times{}^{10}\mathrm{C}_2}{{}^{16}\mathrm{C}_5}\\ &= \frac{20\times 45}{4368}\\ &= \frac{75}{364}\end{aligned}$$

## 2.6 Further Question Types

| Type | Method |
|---|---|
| Range of $\mathrm{P}(A\cap B)$ | Label $\mathrm{P}(A\cap B)=x$, then label all other areas in terms of $x$. Let each area $\geqslant 0$ and solve. |
| Games: "first to win", taking turns | list the winning sequences; the game can go on forever; the probabilities form a **GP**; $S_\infty = \frac{a}{1-r}$ $(\lvert r\rvert<1)$ |
| Venn diagram with unknowns ($x$, $y$) | label the regions (2 or 3 events) with $x$, $y$, …; regions add to 1; use some other info in the equation to link $x$ and $y$; every region $\geqslant 0$ gives the greatest / least value |

**Example 2.3**

$\mathrm{P}(A)=0.45$, $\mathrm{P}(B)=0.7$. Find the range of values of $\mathrm{P}(A\cap B)$.

let $\mathrm{P}(A\cap B)=x$

$\mathrm{P}(A\cap B')=0.45-x$

$\mathrm{P}(A'\cap B)=0.7-x$

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-(0.45-x)-(0.7-x)-x\\ &= x-0.15\end{aligned}$$

$\because$ each area $\geqslant 0$

$\therefore\ 0.15\leqslant \mathrm{P}(A\cap B)\leqslant 0.45$

**Example 2.4**

$A$, $B$ take turns to roll a die, $A$ first; first six wins

$$\begin{aligned}\mathrm{P}(A \text{ wins}) &= \frac{1}{6}+\left(\frac{5}{6}\right)^2\frac{1}{6}+\left(\frac{5}{6}\right)^4\frac{1}{6}+\cdots\\ &= \frac{1/6}{1-25/36}\\ &= \frac{6}{11}\end{aligned}$$

# 3 Discrete Random Variables

## 3.1 Probability Distribution

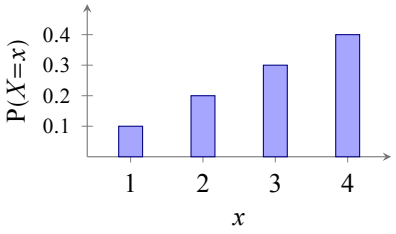
| Item | Result | Note |
|---|---|---|
| Discrete r.v. $X$ | takes separate values $x_1, x_2, \ldots$ (counts) | e.g. number of heads |
| Probability distribution | table of $x$ against $\mathrm{P}(X=x)$ | |
| Conditions | $0 \leqslant \mathrm{P}(X=x) \leqslant 1$ and $\sum_{\text{all } x} \mathrm{P}(X=x) = 1$ | use to find an unknown constant |

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $k$ | $2k$ | $3k$ | $4k$ |

$$\sum \mathrm{P}(X=x) = 1$$

$$10k = 1$$

$$\therefore k = 0.1$$




## 3.2 Expectation and Variance

| Result | Formula | Note |
|---|---|---|
| Expectation (mean) | $\mathrm{E}(X) = \mu = \sum x\,\mathrm{P}(X=x)$ | long-run average |
| Expectation of a function | $\mathrm{E}(g(X)) = \sum g(x)\,\mathrm{P}(X=x)$ | e.g. $\mathrm{E}(X^2) = \sum x^2 \mathrm{P}(X=x)$ |
| Variance | $\mathrm{Var}(X) = \mathrm{E}[(X-\mu)^2] = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$ | $\mathrm{Var}(X) \geqslant 0$ |
| Standard deviation | $\sigma = \sqrt{\mathrm{Var}(X)}$ | |

**GC:** enter $x$ in **L1**, $\mathrm{P}(X=x)$ in **L2**. STAT → **CALC** → **1-Var Stats L1,L2**. Then $\bar{x} = \mathrm{E}(X)$, $\sigma\mathbf{x} = \sigma$ (not **Sx**); $\mathrm{Var}(X) = \sigma^2$. $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$, or read $\Sigma\mathbf{x}^2$.

$$\mathrm{E}(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$$

$$\mathrm{E}(X^2) = 1(0.1) + 4(0.2) + 9(0.3) + 16(0.4) = 10$$

$$\mathrm{Var}(X) = 10 - 3^2 = 1$$

## 3.3 Linear Functions and Combinations

| Result | Formula | Condition |
|---|---|---|
| Linear function | $\mathrm{E}(aX+b)=a\mathrm{E}(X)+b$ | always |
| | $\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$ | always; $b$ does not change spread |
| Sum / difference | $\mathrm{E}(aX\pm bY)=a\mathrm{E}(X)\pm b\mathrm{E}(Y)$ | always |
| | $\mathrm{Var}(aX\pm bY)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$ | $X$, $Y$ **independent**; variances **add** for $\pm$ |
| Sum of $n$ observations | $\mathrm{E}(X_1+X_2+\cdots+X_n)=n\mu$ | always (independence not needed) |
| | $\mathrm{Var}(X_1+X_2+\cdots+X_n)=n\sigma^2$ | $X_1,\ldots,X_n$ **independent** observations of $X$ |
| One observation $\times n$ | $\mathrm{E}(nX)=n\mu$ | always |
| | $\mathrm{Var}(nX)=n^2\sigma^2$ | always; $\mathrm{Var}(nX)\neq\mathrm{Var}(X_1+\cdots+X_n)$ |

**Example 3.1**

$Y=3X-2$, using $\mathrm{E}(X)=3$, $\mathrm{Var}(X)=1$

$$\mathrm{E}(Y)=3(3)-2=7$$

$$\mathrm{Var}(Y)=3^2(1)=9$$

**Example 3.2**

Prize $W$: 0, 4, 10 with probabilities 0.6, 0.3, 0.1; cost to play \$3

$$\mathrm{E}(W)=0+1.2+1=2.2$$

$$\mathrm{E}(\text{gain})=2.2-3=-0.8$$

$\therefore$ not a fair game

**Note**

- If a game is **fair**, $\mathrm{E}(\text{gain})=0$.
- To find the expected value of a variable other than the original $X$, there are two options:
  (1) link the new variable to the old one, e.g. $Y=5-2X$, then apply the expectation formulas;
  (2) draw an entirely new table for the new variable and find the expectation directly from that table.

# 4 Binomial Distribution

## 4.1 The Binomial Model

| Item | Result | Note |
|---|---|---|
| Notation | $X \sim \mathrm{B}(n,p)$ | $X$ = no. of successes in $n$ trials |
| Probability | $\mathrm{P}(X=r) = {}^{n}\mathrm{C}_{r}\, p^{r}(1-p)^{n-r}, \quad r=0,1,\ldots,n$ | MF27 |
| Mean | $\mathrm{E}(X) = np$ | MF27 (no proof) |
| Variance | $\mathrm{Var}(X) = np(1-p)$ | MF27 (no proof) |

**Conditions for $X \sim \mathrm{B}(n,p)$ (state in context)**

- fixed number $n$ of trials
- each trial has exactly 2 outcomes: success / failure
- $\mathrm{P}(\text{success}) = p$ is the **same** for every trial
- trials are **independent**

Note

$X \sim \mathrm{B}(n,p)$ counts the number of successes out of $n$ trials. So $0 \leqslant X \leqslant n$, $X \in \mathbb{Z}$.

| Model fails when … | Why |
|---|---|
| selecting without replacement from a small population | $p$ changes; trials not independent |
| counting trials until the first success | $n$ is not fixed |
| events are linked (e.g. family members, same day) | trials not independent |
| $p$ varies (e.g. by time of day, by type of item) | $p$ not constant |

## 4.2 Using the GC

| Probability | Conversion | GC command |
|---|---|---|
| $\mathrm{P}(X=r)$ | not needed | `binompdf(n,p,r)` |
| $\mathrm{P}(X \leqslant r)$ | not needed | `binomcdf(n,p,r)` |
| $\mathrm{P}(X<r)$ | $\mathrm{P}(X \leqslant r-1)$ | `binomcdf(n,p,r-1)` |
| $\mathrm{P}(X \geqslant r)$ | $1-\mathrm{P}(X \leqslant r-1)$ | `1-binomcdf(n,p,r-1)` |
| $\mathrm{P}(X>r)$ | $1-\mathrm{P}(X \leqslant r)$ | `1-binomcdf(n,p,r)` |
| $\mathrm{P}(a \leqslant X \leqslant b)$ | $\mathrm{P}(X \leqslant b)-\mathrm{P}(X \leqslant a-1)$ | `binomcdf(n,p,b)-binomcdf(n,p,a-1)` |

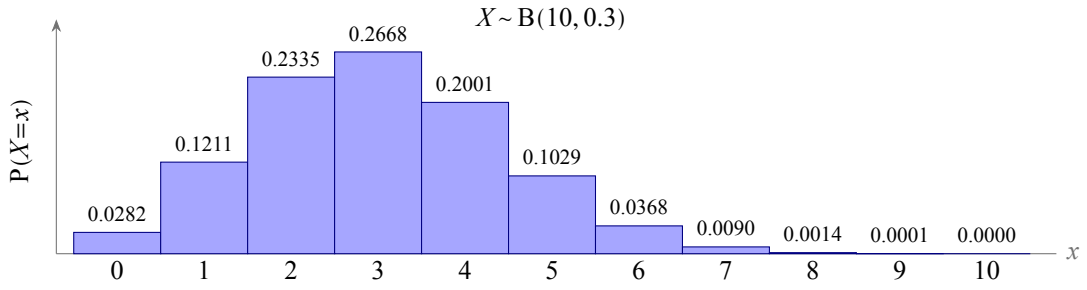
**Note**

- **binompdf(…)**: $P(X=r)$
- **binomcdf(…)**: $P(X \leqslant r)$

Any other inequalities must be converted to the above two forms before keying them into your GC.

**GC:** 2nd VARS (DISTR) → **binompdf(** / **binomcdf(**.

**GC:** solve for $p$ or $n$: Y=, $Y_1 = $ **binomcdf(10,X,3)** (X plays $p$), $Y_2 =$ the given value; GRAPH; 2nd TRACE (CALC) → **5:intersect**. For $n$: use 2nd GRAPH (TABLE) instead.




**Example 4.1**

$X \sim \mathrm{B}(10, 0.3)$

$$\begin{aligned} \mathrm{P}(X=3) &= {}^{10}\mathrm{C}_3 (0.3)^3 (0.7)^7 \\ &= 0.267 \\ \mathrm{P}(X \geqslant 4) &= 1 - \mathrm{P}(X \leqslant 3) \\ &= 1 - 0.6496 = 0.350 \end{aligned}$$

$$\mathrm{E}(X) = 3, \ \mathrm{Var}(X) = 2.1$$

**Example 4.2**

$X \sim \mathrm{B}(20, 0.4)$. Find the probability that $X$ is within 2 standard deviations of the mean.

$$\begin{aligned} \mu &= 20(0.4) = 8 \\ \sigma &= \sqrt{20(0.4)(0.6)} \\ &= \sqrt{4.8} = 2.1909 \\ \mu - 2\sigma &= 3.618 \\ \mu + 2\sigma &= 12.382 \\ \mathrm{P}(3.618 < X < 12.382) &= \mathrm{P}(4 \leqslant X \leqslant 12) \\ &= \mathrm{P}(X \leqslant 12) - \mathrm{P}(X \leqslant 3) \\ &= 0.97897 - 0.01596 \\ &= 0.963 \end{aligned}$$

**Example 4.3**

$X \sim \mathrm{B}(10, 0.3)$. Find the probability that the tenth result is the 3rd success.

$$\text{let } Y \sim \mathrm{B}(9, 0.3)$$

$$\mathrm{P}(Y=2) = {}^{9}\mathrm{C}_2 (0.3)^2 (0.7)^7$$

$$= 0.26683$$

$$\therefore \mathrm{P}(\text{10th is 3rd success}) = \mathrm{P}(Y=2) \times 0.3$$

$$= 0.26683 \times 0.3$$

$$= 0.0800$$

## 4.3 Common Question Types

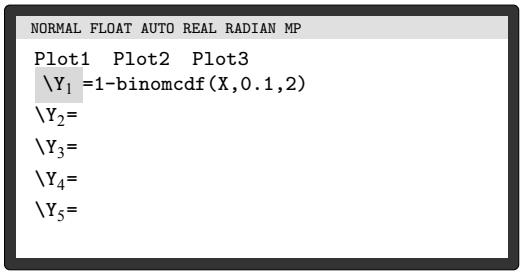
| Type | Method |
|---|---|
| Smallest $n$ with $\mathrm{P}(X \geqslant 1) > c$ | Let $X \sim \mathrm{B}(n, p)$ ($p$ is given), then use **GC table** to solve for $n$. |
| Unknown $p$ | form the equation in $p$ (e.g. $\mathrm{P}(X=0) = (1-p)^n$); solve; or use GC graph to solve for $p$ |
| Given a condition | conditional probability: $\mathrm{P}(X=r \mid X \geqslant 1) = \dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X \geqslant 1)}$, $r \geqslant 1$ |
| Most probable value (mode) $r$ | **Compare mode $r$ with its directly adjacent neighbours**<br>• $\mathrm{P}(X=r) > \mathrm{P}(X=r+1)$<br>• $\mathrm{P}(X=r) > \mathrm{P}(X=r-1)$<br>Equivalently, solve $\dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X=r+1)} > 1$ |
| Trial $r$ is the $k$-th success | first $r-1$ trials contain $k-1$ successes, then a success: ${}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k} \times p$<br>Alternatively, $\underbrace{{}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k}}_{\text{define } Y \sim \mathrm{B}(r-1, p), \text{ and } \mathrm{P}(Y=k-1)} \times p$ |
| Total / mean of $N$ independent copies of $X \sim \mathrm{B}(n, p)$ ($N$ is large) | Use the CLT (Ch 6) with $\mu = np$, $\sigma^2 = np(1-p)$<br>• $\bar{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{N}\right) = \mathrm{N}\left(np, \dfrac{np(1-p)}{N}\right)$ approximately by CLT<br>• $X_1 + X_2 + \cdots + X_N \sim \mathrm{N}(N\mu, N\sigma^2) = \mathrm{N}(Nnp, Nnp(1-p))$ approximately by CLT<br>For use of CLT, usually coupled with the phrases "estimate the probability" or "by using a suitable approximation". |
| Estimate $p$ from data | $p \approx \dfrac{\text{total successes}}{\text{total trials}} = \dfrac{\bar{x}}{n}$ (frequency table: $\sum xf / n \sum f$) |
| Sketch of $\mathrm{B}(n, p)$ | vertical bars at $r = 0, 1, \ldots, n$; label each probability above the bars; skewed right if $p < 0.5$, left if $p > 0.5$, symmetric if $p = 0.5$ |
| Double binomial | new $p'$ from the first part, e.g. $Y \sim \mathrm{B}(m, p')$ where $p' = \mathrm{P}(X \geqslant 2)$ ("on how many of $m$ days") |

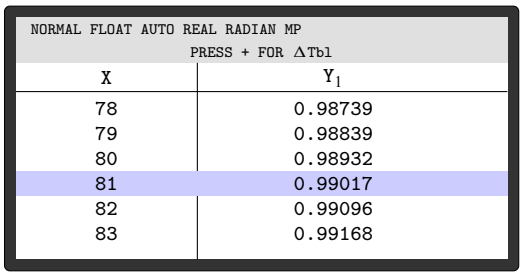
**Example 4.4**

$X \sim \mathrm{B}(n, 0.1)$. Find the smallest value of $n$ such that $\mathrm{P}(X \geqslant 3) > 0.99$.

$$\mathrm{P}(X \geqslant 3) > 0.99$$

$$1 - \mathrm{P}(X \leqslant 2) > 0.99$$




$$\therefore n_{\min} = 81$$

**Example 4.5**

$X \sim \mathrm{B}(10, p)$. Given that the mode of $X$ is 3, find the range of $p$.

$$\mathrm{P}(X=3) > \mathrm{P}(X=2)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_2 p^2 (1-p)^8$$

$$120p > 45(1-p)$$

$$165p > 45$$

$$p > \frac{3}{11}$$

$$\mathrm{P}(X=3) > \mathrm{P}(X=4)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_4 p^4 (1-p)^6$$

$$120(1-p) > 210p$$

$$120 > 330p$$

$$p < \frac{4}{11}$$

$$\therefore \frac{3}{11} < p < \frac{4}{11}$$

**Example 4.6**

The number of defective items in a box of 20 is modelled by $X \sim \mathrm{B}(20, 0.3)$. A random sample of 50 boxes is taken. By using a suitable approximation, estimate the probability that the mean number of defective items per box exceeds 6.2.

$$\mathrm{E}(X) = 20(0.3) = 6$$

$$\mathrm{Var}(X) = 20(0.3)(0.7) = 4.2$$

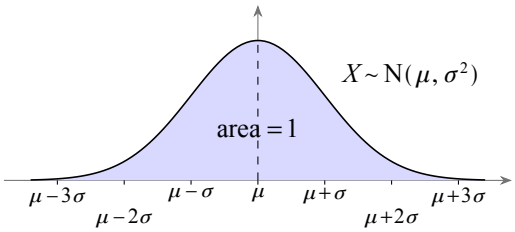
$\because N = 50$ is large,

$$\bar{X} \sim \mathrm{N}\left(6, \frac{4.2}{50}\right) \text{ approximately by CLT}$$

$$\therefore \mathrm{P}(\bar{X} > 6.2) = 0.245$$

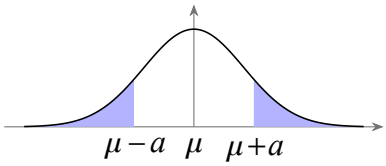
# 5 Normal Distribution

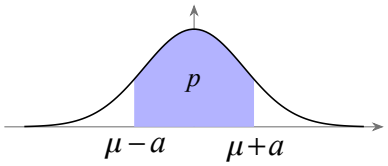
## 5.1 Properties




- bell-shaped, symmetric about $x=\mu$
- total area = 1; mean = $\mu$; variance = $\sigma^2$
- continuous: $\mathrm{P}(X=a)=0$, so $\mathrm{P}(X<a)=\mathrm{P}(X\leqslant a)$
- $\sigma$ larger $\Rightarrow$ wider and flatter

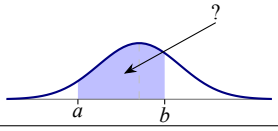
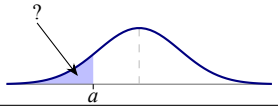
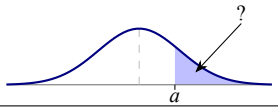
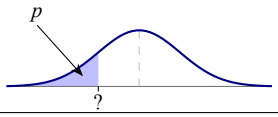
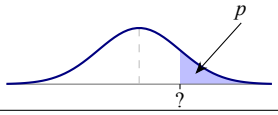
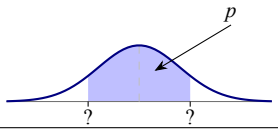
| Item | Result | Note |
|---|---|---|
| Standardise | $Z=\dfrac{X-\mu}{\sigma}\sim \mathrm{N}(0,1)$ | $Z$ = number of s.d. from the mean |
| Symmetry | $\mathrm{P}(X<\mu-a)=\mathrm{P}(X>\mu+a)$ | $\mathrm{P}(Z<-z)=\mathrm{P}(Z>z)=1-\mathrm{P}(Z<z)$ |
| Complement | $\mathrm{P}(X>a)=1-\mathrm{P}(X<a)$ | total area is 1 |




Left: equal tails. Right: $\mathrm{P}(\mu-a<X<\mu+a)=p$, so each tail is $\frac{1-p}{2}$.

## 5.2 Finding Probabilities and Values (GC)

| Find | GC command | Graph |
|---|---|---|
| $P(a<X<b)$ | `normalcdf(a,b,μ,σ)` |   |
| $P(X<a)$ | `normalcdf(-1E99,a,μ,σ)` |   |
| $P(X>a)$ | `normalcdf(a,1E99,μ,σ)` |   |
| $x$ with $P(X<x)=p$ | `invNorm(p,μ,σ,LEFT)` |   |
| $x$ with $P(X>x)=p$ | `invNorm(p,μ,σ,RIGHT)` |   |
| $x$ with $P(\mu-c<X<\mu+c)=p$ | `invNorm(p,μ,σ,CENTER)`<br>the two values $(\mu \pm c)$ need to be symmetrical about $\mu$ |   |

**Note**

Remember to square root your variance before keying into GC!

**Example 5.1**

$X \sim \mathrm{N}(60, 8^2)$: find $\mathrm{P}(52 < X < 70)$, and $x$ with $\mathrm{P}(X > x) = 0.2$

$$\mathrm{P}(52 < X < 70) = 0.736$$

$$\mathrm{P}(X < x) = 0.8$$

$$\therefore x = 66.7$$

**Example 5.2**

Find $\mu$, $\sigma$ given $\mathrm{P}(X < 30) = 0.1$ and $\mathrm{P}(X > 46) = 0.2$

Let $X \sim \mathrm{N}(\mu, \sigma^2)$

$$\mathrm{P}(X < 30) = 0.1, \quad \mathrm{P}(X > 46) = 0.2$$

$$\mathrm{P}\left(Z < \tfrac{30-\mu}{\sigma}\right) = 0.1, \quad \mathrm{P}\left(Z > \tfrac{46-\mu}{\sigma}\right) = 0.2$$

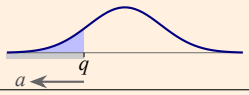
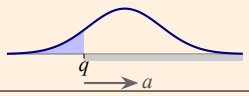
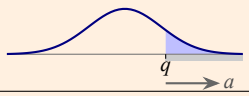
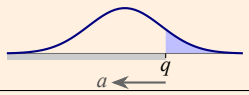
$$\frac{30-\mu}{\sigma} = -1.2816, \quad \frac{46-\mu}{\sigma} = 0.8416$$

Solving simultaneously,

$$\sigma = 7.54, \quad \mu = 39.7$$

Sometimes if the probabilities given are similar, e.g. $\mathrm{P}(X < 30) = 0.1$ & $\mathrm{P}(X > 50) = 0.1$, we should use symmetry to find $\mu$, i.e. $\mu = \frac{30+50}{2} = 40$.

! invNorm Double Inequalities

| Inequality | Solution | Graph | Note |
|---|---|---|---|
| $\mathrm{P}(X \leqslant a) < k$ | $a <$ `invNorm(k,μ,σ,LEFT)` |   | keep the outer sign < |
| $\mathrm{P}(X \leqslant a) > k$ | $a >$ `invNorm(k,μ,σ,LEFT)` |   | keep the outer sign > |
| $\mathrm{P}(X \geqslant a) < k$ | $a >$ `invNorm(k,μ,σ,RIGHT)` |   | flip the outer sign < |
| $\mathrm{P}(X \geqslant a) > k$ | $a <$ `invNorm(k,μ,σ,RIGHT)` |   | flip the outer sign > |

## 5.3 Linear Functions and Combinations of Normals

| Situation | Distribution | Condition |
|---|---|---|
| $aX+b$ | $\mathrm{N}(a\mu+b, a^2\sigma^2)$ | $X$ normal |
| $aX \pm bY$ | $\mathrm{N}(a\mu_X \pm b\mu_Y, a^2\sigma_X^2+b^2\sigma_Y^2)$ | $X$, $Y$ independent normals |
| $X_1+X_2+\cdots+X_n$ | $\mathrm{N}(n\mu, n\sigma^2)$ | independent copies of $X$ |
| $nX$ | $\mathrm{N}(n\mu, n^2\sigma^2)$ | one observation × $n$ |

**Common mistake**

- Variances **add** for both $X+Y$ and $X-Y$: $\text{Var}(X-Y)=\text{Var}(X)+\text{Var}(Y)$.
- $\text{Var}(2X)=4\text{Var}(X)\neq\text{Var}(X_1+X_2)=2\text{Var}(X)$.

**Example 5.3**

$X\sim\text{N}(120,15^2)$, $Y\sim\text{N}(100,8^2)$ independent; $\text{P}(X>Y)$

$$X-Y\sim\text{N}(20,15^2+8^2)$$
$$=\text{N}(20,17^2)$$
$$\text{P}(X>Y)=\text{P}(X-Y>0)$$
$$=0.880$$

## 5.4 Further Question Types

| Type | Method |
|---|---|
| Percentage change ("loses 5%") | new variable is $0.95X$: $\text{N}(0.95\mu,0.95^2\sigma^2)$ (not $X-5$) |
| Total of different items (e.g. 2 of $X$ and 3 of $Y$) | $T=X_1+X_2+Y_1+Y_2+Y_3$: $\text{E}(T)=2\mu_X+3\mu_Y$, $\text{Var}(T)=2\sigma_X^2+3\sigma_Y^2$; with scaled items add $a^2\sigma^2$ |
| "differs by less than $c$" | $\text{P}(\lvert X-Y\rvert<c)=\text{P}(-c<X-Y<c)$: `normalcdf(-c,c,μ,σ)` |
| Unknown $\mu$ or $\sigma$ (one condition) | standardise: $\frac{a-\mu}{\sigma}=$ `invNorm(p)`, solve (the sign of $z$ follows the tail) |
| Range of $\sigma$ (or $\mu$) for $\text{P}(X>a)<p$ | $\frac{a-\mu}{\sigma}>$ `invNorm(1-p)`; $\sigma>0$, so multiply through without reversing; then solve |
| Normal event, then count ("$k$ of $n$ items") | $p=\text{P}(\text{event})$ by GC; then $\text{B}(n,p)$ |
| Sketch of normal curves | bell shape, axis marked at $\mu$, $\mu\pm\sigma$, $\mu\pm2\sigma$, $\mu\pm3\sigma$; same $\mu$: larger $\sigma$ is wider and lower; different $\mu$: curves shifted; each has area 1 |
| Model unsuitable | compute $\text{P}(X<0)$ (or outside a natural range): not negligible; or non-symmetric data |

**Example 5.4**

$X\sim\text{N}(50,3^2)$, $Y\sim\text{N}(30,2^2)$ independent; $T=2X+3Y$, find $\text{P}(T>200)$

$$\text{E}(T)=2(50)+3(30)=190$$
$$\text{Var}(T)=4(9)+9(4)=72$$
$$T\sim\text{N}(190,72)$$
$$\text{P}(T>200)=0.119$$

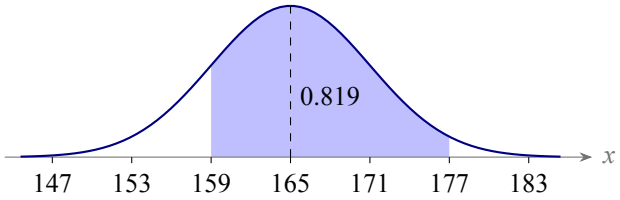
**Example 5.5**

The mass, $X$ grams, of a bag of rice is modelled by $X\sim\text{N}(500,12^2)$. A bag is underweight if its mass is less than 480 grams. 20 bags are chosen at random. Find the probability that at least 2 of the bags are underweight.

$$X\sim\text{N}(500,12^2)$$
$$\text{P}(X<480)=0.04779$$
$$\text{let } Y\sim\text{B}(20,0.04779)$$
$$\text{P}(Y\geqslant2)=1-\text{P}(Y\leqslant1)$$
$$=1-0.75249$$
$$=0.248$$

**Example 5.6**

The height, $X$ cm, of a student is modelled by $X \sim \mathrm{N}(165, 6^2)$. Sketch the normal curve of $X$, marking $\mu \pm \sigma$, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$ on the horizontal axis, and shade the region representing $\mathrm{P}(159 < X < 177)$. Hence find $\mathrm{P}(159 < X < 177)$.




$$\begin{aligned}\mathrm{P}(159 < X < 177) &= \mathrm{P}(-1 < Z < 2) \\ &= 0.819\end{aligned}$$

# 6 Sampling and the Central Limit Theorem

## 6.1 Population and Sample

| Term | Meaning | Notation |
|---|---|---|
| Population | the whole group of interest | mean $\mu$, variance $\sigma^2$ (parameters) |
| Sample | a part of the population, size $n$ | $\bar{x}$, $s^2$ (statistics) |
| Random sample | • every member of the population is equally likely to be chosen<br>• the selection is independent<br>**Benefits:** avoid bias, and ensure the sample is representative of the population.<br>**How to obtain:** number every member of the population with a distinct positive integer from 1 to $N$. Use a computer to randomly generate $n$ distinct positive integers. Select the members whose numbers correspond to those generated by the computer. | |

| Not random if … | Reason |
|---|---|
| chosen by convenience (e.g. first 30 in the canteen, volunteers) | some members are more likely than others |
| only part of the population can be chosen | the rest has no chance |
| one choice affects another (e.g. friends chosen together) | not independent |

## 6.2 Unbiased Estimates

| Estimate of | Raw data | Coded: $\Sigma(x-a)$, $\Sigma(x-a)^2$ |
|---|---|---|
| Mean $\mu$ | $\bar{x}=\dfrac{\sum x}{n}$ | $\bar{x}=a+\dfrac{\sum(x-a)}{n}$ |
| Variance $\sigma^2$ | $s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$<br>or $s^2=\dfrac{1}{n-1}\sum(x-\bar{x})^2$ | $s^2=\dfrac{1}{n-1}\left[\sum(x-a)^2-\dfrac{\left(\sum(x-a)\right)^2}{n}\right]$ |

**GC:** STAT → CALC → 1-Var Stats. Use Sx (=$s$, divides by $n-1$), so $s^2=$ Sx$^2$. For a frequency table, enter the values in L1 and the frequencies in L2, then 1-Var Stats L1,L2. Do **not** use $\sigma$x (divides by $n$; this is not the unbiased estimate).

**Example 6.1**

$n=10, \sum x=125, \sum x^2=1830$

$$\bar{x}=\frac{125}{10}=12.5$$

$$s^2=\frac{1}{9}\left[1830-\frac{125^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

**Example 6.2**

given $\sum(x-10)=25, \sum(x-10)^2=330$

$$\bar{x}=10+\frac{25}{10}=12.5$$

$$s^2=\frac{1}{9}\left[330-\frac{25^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

**Unbiased estimate of population variance**

$$s^2=\frac{n}{n-1}\sigma_x^2$$

where $\sigma_x^2=\frac{1}{n}\sum(x-\bar{x})^2$ is the **sample variance** (GC: $\sigma x^2$).

**Example 6.3**

(Hypothesis testing) [VJC 2025 Prelim P2 Q12]

A hospital introduces a new medication regimen to help patients recover from a viral infection more quickly. Previously, the average recovery time was 7.2 days.

…

It was discovered that an intern at the hospital had made an error in recording the recovery times. A random sample of another 30 patients was taken and their recovery times under the new regimen was taken and recorded by a senior researcher. The sample standard deviation was found to be 1.7 days and the sample mean time was denoted by $k$.

(f) A test at the 1% significance level concludes that the average recovery time did not improve from 7.2 days. Find the set of values of $k$. [3]

$$s^2=\frac{n}{n-1}(\text{sample variance})$$

$$=\frac{30}{29}(1.7)^2$$

$$=2.9897$$

**Example 6.4**

The masses, in kg, of a random sample of 5 parcels are 4, 7, 8, 10, 11. Find unbiased estimates of the population mean and variance.

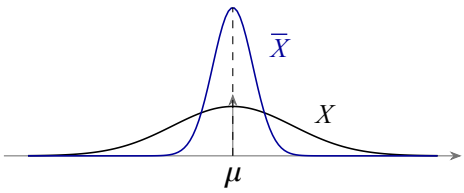
$$\bar{x} = \frac{4+7+8+10+11}{5} = 8$$

$$\sum (x-\bar{x})^2 = (-4)^2+(-1)^2+0^2+2^2+3^2$$
$$= 30$$

$$s^2 = \frac{1}{n-1}\sum (x-\bar{x})^2$$
$$= \frac{30}{4}$$
$$= 7.5$$

## 6.3 Distribution of the Sample Mean

| Population | Distribution of $\overline{X}$ | Reason |
|---|---|---|
| Normal, $X \sim \mathrm{N}(\mu, \sigma^2)$ | $\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ (exact) | any $n$; linear combination of normals |
| Not normal / unknown, $n$ **large** (e.g. $n \geqslant 30$) | $\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ approximately by CLT | **Central Limit Theorem** |




Same centre $\mu$. $\overline{X}$ is **narrower**: s.d. $\frac{\sigma}{\sqrt{n}}$ (larger $n$, narrower).

**Example 6.5**

population mean 80, s.d. 12, $n = 40$

$$\overline{X} \sim \mathrm{N}(80, \frac{12^2}{40}) \text{ approximately by CLT}$$

$$\mathrm{P}(\overline{X} > 83) = 0.0569$$

**Central Limit Theorem**

For a random sample of size $n$ (large, e.g. $n \geqslant 30$) from **any** population with mean $\mu$ and variance $\sigma^2$:

- $\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ approximately by CLT.
- $X_1 + X_2 + \cdots + X_n \sim \mathrm{N}(n\mu, n\sigma^2)$ approximately by CLT.

## 6.4 Which Distribution?

| The question … | Use | Key step |
|---|---|---|
| number of successes in $n$ independent trials | $\mathrm{B}(n, p)$ | state the conditions |
| one measurement, normal | $\mathrm{N}(\mu, \sigma^2)$ | standardise / GC |
| total or difference of independent normals | $\mathrm{N}(\cdot, \cdot)$ | add the **variances** |
| mean of $n$ values, population normal | $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ | exact |
| mean of $n$ values, population not normal, $n$ large | $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ approximately by CLT | say "by CLT" |

## 6.5 Further Question Types

| Type | Method |
|---|---|
| Population given by a table or $\mathrm{B}(n, p)$, $n$ observations large | $\mu = \mathrm{E}(X)$, $\sigma^2 = \mathrm{Var}(X)$ first (Ch 3, 4); then $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ approximately by CLT |
| Total of many independent counts (e.g. defects in 100 batches) | $S = X_1 + \cdots + X_n \sim \mathrm{N}(n\mu, n\sigma^2)$ approximately by CLT |
| Least $n$ for $\mathrm{P}(\overline{X} < c) \geqslant p$ | $\frac{c-\mu}{\sigma/\sqrt{n}} \geqslant$ `invNorm(p)`, solve for $\sqrt{n}$; or GC table: $\mathrm{Y}_1 =$ `normalcdf` with s.d. $\sigma/\sqrt{\mathrm{X}}$, X as $n$ |
| Is the sample size big enough? | $n < 30$ and population not known to be normal: CLT **cannot** be used; $n \geqslant 30$: say "by CLT" |

**Example 6.6**

The number of defective items, $X$, in a box of 12 items is modelled by $X \sim \mathrm{B}(12, 0.25)$. A random sample of 40 boxes is taken. Find the probability that the mean number of defective items per box exceeds 3.4.

$$\mathrm{E}(X) = 12(0.25) = 3$$

$$\mathrm{Var}(X) = 12(0.25)(0.75) = 2.25$$

$\because n = 40$ is large

$$\overline{X} \sim \mathrm{N}(3, \frac{2.25}{40})$$

approximately by CLT

$$\mathrm{P}(\overline{X} > 3.4) = 0.0458$$

**Example 6.7**

The number of faults, $X$, recorded by a sensor in an hour has the following probability distribution.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | 0.4 | 0.3 | 0.2 | 0.1 |

45 independent hourly readings are taken. Find the probability that the total number of faults recorded exceeds 50.

$$\mathrm{E}(X) = 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1)$$

$$= 1$$

$$\mathrm{E}(X^2) = 0(0.4) + 1(0.3) + 4(0.2) + 9(0.1)$$

$$= 2$$

$$\mathrm{Var}(X) = 2 - 1^2 = 1$$

let $S = X_1 + X_2 + \cdots + X_{45}$

$\because n = 45$ is large

$$S \sim \mathrm{N}(45(1), 45(1))$$

approximately by CLT

$$S \sim \mathrm{N}(45, 45)$$

approximately by CLT

$$\mathrm{P}(S > 50) = 0.228$$

# 7 Hypothesis Testing

## 7.1 Terminology

| Term | Meaning |
|---|---|
| Null hypothesis $H_0$ | the claim assumed true: $H_0: \mu = \mu_0$ |
| Alternative hypothesis $H_1$ | what we look for evidence of: $\mu < \mu_0$, $\mu > \mu_0$ (1-tail) or $\mu \neq \mu_0$ (2-tail) |
| Test statistic | $\bar{X}$ (or $Z$), whose distribution is found assuming $H_0$ is true |
| Level of significance $\alpha$ | probability of rejecting $H_0$ when $H_0$ is true; given in the question |
| Critical region | values of the test statistic for which $H_0$ is rejected; boundary = critical value |
| $p$-value | probability, assuming $H_0$ true, of a test statistic at least as extreme as the one observed; smallest level at which $H_0$ is rejected |

## 7.2 Formulating the Hypotheses

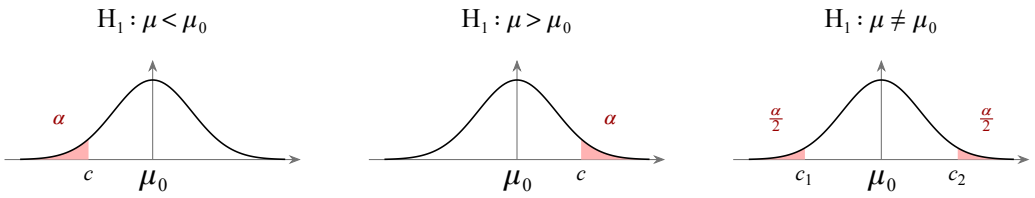
| Wording | $H_1$ | Type |
|---|---|---|
| decreased, less than, lower, overstated | $\mu < \mu_0$ | 1-tail (lower) |
| increased, more than, greater, understated | $\mu > \mu_0$ | 1-tail (upper) |
| changed, different, not correct, biased | $\mu \neq \mu_0$ | 2-tail |

Define $\mu$ in context (e.g. "$\mu$ is the population mean mass"). $H_0$ always uses "=".

## 7.3 Test Statistic and Its Distribution

| Situation | Distribution under $H_0$ | Say |
|---|---|---|
| Normal population, $\sigma^2$ known | $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ | any $n$ |
| Any population, $n$ large, $\sigma^2$ known | $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ approximately by CLT | "by CLT" |
| Any population, $n$ large, $\sigma^2$ unknown | $\bar{X} \sim N(\mu_0, \frac{s^2}{n})$ approximately by CLT | "by CLT"; $s^2$ unbiased estimate |

## 7.4 The Test: *p*-value and Critical Region




| | **Lower tail** | **Upper tail** | **2-tail** |
|---|---|---|---|
| *p*-value | $P(\bar{X} \leqslant \bar{x})$ | $P(\bar{X} \geqslant \bar{x})$ | $2P(\bar{X} \leqslant \bar{x})$ if $\bar{x} < \mu_0$;<br>$2P(\bar{X} \geqslant \bar{x})$ if $\bar{x} > \mu_0$ |
| Reject $H_0$ if | *p*-value $\leqslant \alpha$ | | |
| Critical region | $\bar{X} \leqslant c$, $P(\bar{X} \leqslant c) = \alpha$ | $\bar{X} \geqslant c$, $P(\bar{X} \geqslant c) = \alpha$ | $\bar{X} \leqslant c_1$ or $\bar{X} \geqslant c_2$; each tail $\frac{\alpha}{2}$ |
| Reject $H_0$ if | $\bar{x}$ lies in the critical region | | |

**_p_-value of a two-tailed test**

*p*-value (two-tailed test) = 2 × *p*-value (one-tailed test)

**Steps of a test**

1 $H_0: \mu = \mu_0$, $H_1: \mu \square \mu_0$; define $\mu$; state $\alpha$
2 distribution of $\bar{X}$ under $H_0$ (with reason: normal population / CLT)
3 *p*-value from the GC (or critical region)
4 compare: $p \leqslant \alpha \Rightarrow$ reject $H_0$; $p > \alpha \Rightarrow$ do not reject $H_0$
5 conclusion **in context**

| **Result** | **Conclusion (in context)** |
|---|---|
| Reject $H_0$ | there is sufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |
| Do not reject $H_0$ | there is insufficient evidence, at the $\alpha$ level of significance, to conclude that [$H_1$ in words] |

Never write "accept $H_0$" or "$H_0$ is true / proven".

**GC:** STAT → TESTS → 1:Z-Test. Inpt:Stats; enter $\mu 0$, $\sigma$ (use $s$ if $\sigma$ unknown, large $n$), $\bar{x}$, n; choose $\neq \mu 0$, $< \mu 0$ or $> \mu 0$ to match $H_1$; Calculate gives $z$ and $p$. With raw data use Inpt:Data with the list; for a frequency table first run 1-Var Stats L1,L2 to get $\bar{x}$ and Sx, then use Inpt:Stats.

**Example 7.1**

[N2016/H1/10]

A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1.

(i) Test the scientist's claim at the 5% significance level. [4]

Let $X$ denote the top speed of a randomly selected cheetah in km/h, with population mean $\mu$ km/h.

$H_0: \mu = 95$

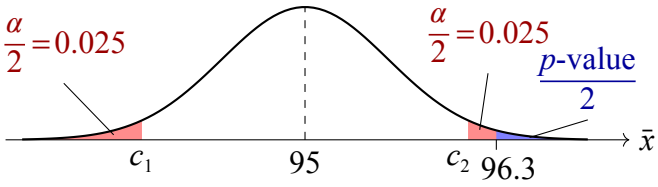
$H_1: \mu \neq 95$

Conduct a 2-tailed test at the 5% level of significance.

Test statistic:

Under $H_0$, *(since $n = 40$ is large, by central limit theorem,)*

$\overline{X} \sim N(95, \frac{4.1^2}{40})$ *approximately*




Reject $H_0$ if $p$-value $\leqslant$ 5% level of significance.

By GC, $p$-value = 0.0449

Thus, since $p$-value < 5%, we reject $H_0$ at the 5% level of significance. There is sufficient evidence to conclude that the mean top speed of cheetahs is not 95km/h. So the scientist's claim is not valid.

## Common Definitions

| Term | Template | Definition (in context) |
|---|---|---|
| Random sample | every member of the population is equally likely to be chosen and the selections are independent. | each cheetah is equally likely to be chosen and the selections are independent |
| Level of significance | there is a probability of $\alpha$% that we wrongly reject $H_0$, when $H_0$ is in fact true. | there is a probability of 5% that we wrongly conclude that the population mean top speed of cheetahs is not 95km/h when it is actually 95km/h. |
| $p$-value | it is the probability that the test statistic is as extreme or more extreme than the observed value given that $H_0$ is true.<br>**OR**<br>It is the smallest level of significance such that $H_0$ is rejected. | it is 2 times the probability that the observed sample mean top speed of cheetahs is more than 96.3km/h when the population mean top speed is 95km/h.<br>**OR**<br>It is the smallest level of significance to conclude that the mean top speed of cheetahs is not 95km/h. |
| Unbiased estimate | it is an estimate whose expected value is equal to the population parameter it is estimating. | |

## 7.5 Using the Result of a Test to Find Unknowns

| Given | Method |
|---|---|
| $H_0$ rejected at level $\alpha$; find range of $\alpha$ | $\alpha \geqslant p$-value (smallest level of significance $= p$-value) |
| Range of $\bar{x}$ (or $\mu_0$, $n$) for which $H_0$ is rejected | critical region: find $c$ by **invNorm**, then write the inequality in $\bar{x}$ |
| Range of $n$ | $p$-value (or $z$) in terms of $n$; use a GC table in $n$ to find where the condition holds |
| $p$-value given; find $\sigma$ (or $\bar{x}$, $n$) | one-tail: $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} =$ **invNorm(1-p)** (upper) or **invNorm(p)** (lower); two-tail: use $\frac{p}{2}$ in the tail |
| One more data value $h$ added to the sample | new mean $= \frac{n\bar{x}+h}{n+1}$; apply the critical region (or $p$-value condition) and solve the inequality for $h$ |
| Values of $\bar{x}$ for which $H_0$ is **not** rejected | the region **between** the critical values (2-tail), or on the other side of $c$ (1-tail) |
| Effect of changing $\sigma$, $n$ or $\alpha$ | larger $\sigma$ (or smaller $n$): $\sigma/\sqrt{n}$ larger, $p$-value larger, harder to reject; smaller $\sigma$ (or larger $n$): easier to reject; larger $\alpha$: easier to reject |
| Assumption needed (small $n$) | population normal; and $\sigma^2$ unchanged / known (state which). Large $n$: none, by CLT |
| Sample not suitable for the test | sample not random (bias), or $n$ too small for CLT when the population is not known to be normal |
| One-tail or two-tail? | decided by the wording **before** looking at the data: a direction ("reduced", "higher") ⇒ 1-tail; "changed" / "correct?" ⇒ 2-tail |

**Example 7.2**

$H_1 : \mu > 50;\ \sigma = 8,\ n = 40,\ \alpha = 5\%$. Critical region?

$$\overline{X} \sim \mathrm{N}(50, \frac{8^2}{40})$$

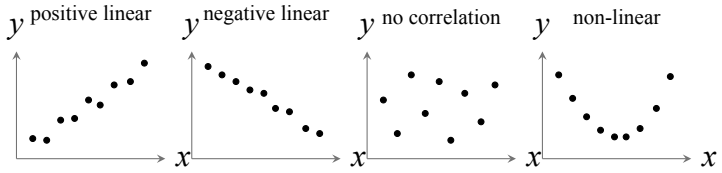
approximately by CLT

$$c = \mathrm{invNorm}(0.95, 50, \sqrt{1.6})$$
$$= 52.08$$

$\therefore$ reject if $\bar{x} \geqslant 52.08$

# 8 Correlation and Linear Regression

## 8.1 Scatter Diagrams




| Item | Note |
|---|---|
| Independent $x$, dependent $y$ | $x$: controlled / explains; $y$: the response (e.g. $y$ = sales, $x$ = advertising) |
| Describe a scatter diagram | direction (positive / negative), linear or curved, strength (close to a line), outliers |
| Outlier | point far from the pattern; check data, comment on its effect on $r$ and on the line |
| GC | enter $x$ in **L1**, $y$ in **L2**; 2nd Y= (STAT PLOT) → **Plot1** on, scatter; ZOOM **9:ZoomStat** |

## 8.2 Product Moment Correlation Coefficient $r$

**PMCC (MF27)**

$$r=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx}=\sum x^2-\frac{\left(\sum x\right)^2}{n},\quad S_{yy}=\sum y^2-\frac{\left(\sum y\right)^2}{n},\quad S_{xy}=\sum xy-\frac{\sum x\sum y}{n}$$

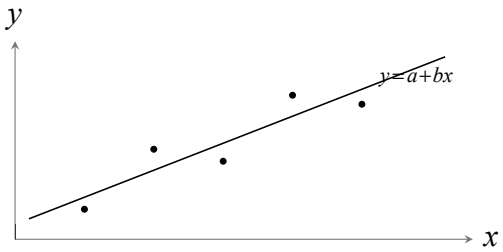
| Value of $r$ | Interpretation |
|---|---|
| $r=1$ or $-1$ | all points lie exactly on a straight line (positive / negative gradient) |
| $r$ close to 1 | strong positive linear correlation |
| $r$ close to $-1$ | strong negative linear correlation |
| $r$ close to 0 | little or no **linear** correlation (a non-linear relation may still exist) |

Note

$-1\leqslant r\leqslant 1$. $|r|$ close to 1 only says a **linear** model fits well; check the scatter diagram. A strong correlation does **not** mean that $x$ causes $y$.

**GC:** 2nd 0 (CATALOG) → **DiagnosticOn** (once). Then STAT → **CALC** → **4:LinReg(ax+b) L1,L2** gives **a**, **b**, **r**. Use **8:LinReg(a+bx)** for $y=a+bx$.

## 8.3 Least-Squares Regression Line




Line of $y$ on $x$ minimises $\sum$(vertical distances)$^2$ (red).

| | $y$ **on** $x$ | $x$ **on** $y$ |
|---|---|---|
| Equation | $y-\bar{y}=b(x-\bar{x}),\ b=\frac{S_{xy}}{S_{xx}}$ | $x-\bar{x}=d(y-\bar{y}),\ d=\frac{S_{xy}}{S_{yy}}$ |
| Or | $y=a+bx,\ a=\bar{y}-b\bar{x}$ | $x=c+dy,\ c=\bar{x}-d\bar{y}$ |
| Use to estimate | $y$ from a given $x$ | $x$ from a given $y$ |

**Note**

Both lines pass through $(\bar{x},\bar{y})$. To estimate $x$ from $y$, use the line of $x$ on $y$ (do **not** rearrange $y$ on $x$), whichever variable is the independent one. The gradient $b$ has the sign of $r$.

**Example 8.1**

$x$: 1, 2, 3, 4, 5, 6 $y$: 3.1, 4.3, 5.2, 6.8, 7.1, 8.9

$$\sum x=21,\ \sum y=35.4,\ \sum xy=143.4,\ \sum x^2=91,\ \sum y^2=231$$

$$S_{xx}=91-\frac{21^2}{6}=17.5$$

$$S_{xy}=143.4-\frac{21(35.4)}{6}=19.5$$

$$S_{yy}=231-\frac{35.4^2}{6}=22.14$$

$$r=\frac{19.5}{\sqrt{17.5\times 22.14}}=0.991$$

$$b=\frac{19.5}{17.5}=1.114,\quad a=5.9-1.114(3.5)=2.00$$

$$\therefore y=2.00+1.11x$$

## 8.4 Interpolation, Extrapolation and Interpretation

| Item | Note |
|---|---|
| Interpolation | estimate for $x$ **within** the data range: reliable if $\|r\|$ is close to 1 and the scatter diagram is linear |
| Extrapolation | estimate for $x$ **outside** the data range: unreliable, the linear relation may not continue |
| Gradient $b$ | the change in $y$ per unit increase in $x$ (in context) |
| Intercept $a$ | value of $y$ when $x=0$; meaningful only if $x=0$ is sensible in context |
| How well does the model fit? | $\|r\|$ close to 1 and points close to the line ⇒ good fit; comment on the range of $x$ |

**Example 8.2**

using the example above: $x=3.5$ and $x=10$

$$x=3.5: \ y=2.00+1.114(3.5)=5.90 \ \text{(interpolation, reliable)}$$

$$x=10: \ y=2.00+1.114(10)=13.1 \ \text{(extrapolation, unreliable)}$$

## 8.5 Transformations to Achieve Linearity

Choose the model whose scatter diagram of (transformed variable, $y$) is the most linear, and whose $|r|$ is closest to 1.

| Model | Scatter diagram | Plot | Regression | Shape |
|---|---|---|---|---|
| $y=a+bx^2$ |  | $y$ against $x^2$ | $y$ on $x^2$ | increasing, curving up |
| $y=a+b\ln x$ |  | $y$ against $\ln x$ | $y$ on $\ln x$ | increasing, curving down |
| $y=a+\frac{b}{x}$ |  | $y$ against $\frac{1}{x}$ | $y$ on $\frac{1}{x}$ | decreasing, levelling off |
| $y=a\,e^{bx}$ |  | $\ln y$ against $x$ | $\ln y$ on $x$ | increasing, curving up faster |

Let $w$ be the transformed variable (e.g. $w=x^2$, $\ln x$, $\frac{1}{x}$). Find $r$ and the line of $y$ on $w$ with the GC (new list), then substitute $w=\ldots$ to estimate $y$. For $\ln y$ on $x$: $\ln y=a+bx \Rightarrow y=e^a e^{bx}$.

## 8.6 Further Question Types

| Type | Method |
|---|---|
| Residual and least squares | residual $e = y - (a+bx)$ (observed − predicted); positive above the line. For the least-squares line $\sum e = 0$, so **squares** are used; the line minimising $\sum e^2$ is the best fit; smaller $\sum e^2 \Rightarrow$ better fit |
| Sum of squares of residuals | compute each $e$, square, add (or use the GC); compare two lines or two models by their $\sum e^2$ |
| Missing data value / unknown | the line passes through $(\bar{x}, \bar{y})$: $\bar{y} = a + b\bar{x}$ with the unknown in $\sum x$ or $\sum y$; solve |
| Extra point that leaves the line unchanged | the point $(\bar{x}, \bar{y})$ (the mean point) |
| Change of units or coding, $x' = px+q$ | $r$ unchanged if $p>0$ (sign of $r$ reverses if $p<0$); new line: substitute $x = \frac{x'-q}{p}$ into the fitted equation |
| Remove or change one point | recompute $r$, $a$, $b$ on the GC; comment on the effect (e.g. $\lvert r \rvert$ nearer 1) |
| Choosing a model | scatter shape **in context** (e.g. levels off, cannot be negative), then $\lvert r \rvert$ closest to 1; state the constants and $r$ for the chosen model |
| Estimate $x$ from $y$ (transformed model) | solve the fitted equation for $x$ (e.g. $y = a + \frac{b}{x} \Rightarrow x = \frac{b}{y-a}$), or GC graph intersect |
| Sketch a scatter diagram | label axes with scales; plot every point (label any special point); do not join; show the shape (curve, outlier) |

**GC:** `LinReg(a+bx) L1,L2,Y1` ([VARS] → Y-VARS → `1:Function` → `Y1` stores the line). In `L3` enter `L2-Y1(L1)`. Then [2nd] [STAT] (LIST) → MATH → `5:sum( L3²)` gives $\sum e^2$. Transformed variable: highlight the `L3` header and enter e.g. `ln(L1)`, `L1²` or `1/L1`, then run `LinReg` with `L3,L2`.

**Example 8.3**

$x$: 1, 2, 3, 4 $y$: 3, 6, 6, 10; line $y = 2x+1$

$$e = 0,\ 1,\ -1,\ 1$$

$$\sum e^2 = 0+1+1+1 = 3$$

least squares $y = 1 + 2.1x$

$$\sum e^2 = 2.70 < 3$$

**Example 8.4**

$s = 1.2t + 5$, $t$ in °C; use $F$ in °F

$$t = \frac{5}{9}(F-32)$$

$$s = 1.2 \times \frac{5}{9}(F-32) + 5$$

$$= \frac{2}{3}F - 16.3$$